Bank IQ – Viva-Voce Preparation

22MDC56 Machine Learning Laboratory · likely questions with crisp answers

How to use: read each answer once, then practise saying it in two sentences with one number attached. If you do not know, say what you would measure to find out.

A. Problem, data and preprocessing

Q. What is the project in one sentence?

A classical machine-learning intelligence layer on top of Vasool Raja, our rule-based Regulatory Digital Twin: it classifies every statement line into ten categories better than the regexes, segments the customer, forecasts next month's spend, flags unusual debits and writes business insights — deployed as endpoints and a dashboard, with the rule engine still the only thing that files a claim.

Q. What is Vasool Raja and what did you change in it?

Our HackVerse hackathon project: parsers for CSV/PDF/passbook photo, a digital twin of the account, 19 RBI rules evaluated at their effective dates, findings with evidence chains, a guardian approval flow and a Tamil/English app. Bank IQ changes nothing in it except mounting a new router and adding an Intelligence screen; all ML code is in the new package bankiq/.

Q. Why a synthetic dataset?

Bank statements are personal data; no public labelled dataset of Indian narrations exists. The generator makes 1,27,604 lines for 500 customers × 12 months from five archetypes with seasonality, long-tail templates and noise. The label is the generating category, never a function of the narration string, so the regex baseline can be measured honestly — and it scores only 71.5%.

Q. What are the ten categories?

UPI payment, card/POS, ATM withdrawal, transfer out, loan EMI, bank charge, income credit, transfer in, refund/reversal, interest credit.

Q. How is a narration turned into features?

Upper-case, split letter–digit boundaries (JAN26 → JAN 26), mask every run of 2+ digits as NUM, collapse separators; then word TF-IDF over 1–2-grams and character-within-word TF-IDF over 3–5-grams, plus nine numeric features (amount, log amount, is_debit, is_round, day, weekday, month-end, has_paise, share of the month). 2,789 sparse features; SVD-200 for kNN and boosting.

Q. Why character n-grams?

Narrations are abbreviated, truncated and noisy (CHRG, WDL, RVSL, OCR errors). Character 3–5-grams make 'LOAN INST', 'NACH DR' and 'ACH D-' look alike and survive truncation; this is Cavnar and Trenkle's argument.

Q. How do you avoid leakage?

Vectorisers and the scaler are fitted on the training split only; cross-validation folds are inside the training split; the test split is scored once. For regression the folds are grouped by customer so a customer's months never appear on both sides. For clustering the hidden archetype is never a feature.

Q. What is the customer-month panel?

One row per customer and month: spend, previous spend, income, credits, counts, balances, per-category amounts and counts, calendar month of the month being predicted (integer + sin/cos) — 34 features; the target is next month's spend.

B. Classification (concept 2)

Q. Which classifiers and how did they do?

Rule baseline 71.5% (macro-F1 0.794); Complement Naive Bayes 93.4%; kNN 97.4%; Decision Tree 96.7%; Logistic Regression 99.9% (macro-F1 0.997) — deployed.

Q. Why Complement Naive Bayes rather than Gaussian?

Features are TF-IDF counts, not Gaussian measurements, and the classes are imbalanced. Complement NB estimates each class from the complement of its data, which Rennie et al. showed fixes the bias toward large classes. It runs only on the non-negative text block.

Q. How does the decision tree choose a split?

The feature and threshold that most reduce Gini impurity of the children; depth 30 with class_weight balanced. The depth sweep shows training F1 rising with depth while cross-validated F1 plateaus — over-fitting.

Q. How did you make kNN work on text?

Euclidean distance in 2,800 sparse dimensions is uninformative, so we project to SVD-200 and use cosine distance with distance weighting; there kNN reaches 97%. The k sweep over {1, 3, 7, 15, 31} shows small k over-fitting and large k over-smoothing. Naive Bayes is the weakest learner: its independence assumption cannot combine word, character and numeric evidence.

Q. What is macro-F1 and why use it?

The unweighted mean of per-class F1. Accuracy would reward ignoring rare classes (bank charge, interest credit) that matter most to the product.

Q. Where exactly does ML beat the rules?

Per-class F1: transfer out 0.427 → 1.000, loan EMI 0.783 → 0.985, UPI 0.785 → 0.999. The rules are strongest on charges (0.888) and refunds — what they were written for — and weakest on everything else; the model lifts every class above 0.98. On the long-tail sample statement (merchant-only lines, bank codes) they disagree on 30 of 41 lines and the model is the one that is right; on the rule-friendly samples they agree on 99%.

Q. How do you know the model works on real statements?

The app's sample statements (94 lines in real HDFC/Canara/SBI/Indian Bank formats, written for the regex engine) are a tiny out-of-distribution check with the rule category as reference. It caught a real flaw: with numeric features at full weight the logistic model agreed with the rules on only 83.0% of lines — it had learned amount habits of the generator; at weight 0.1 agreement is 98.9% with no loss on the synthetic test split.

Q. What is the direction mask?

At inference a credit line may receive only a credit category and a debit line only a debit category: probabilities of impossible classes are zeroed and renormalised. A business constraint that removes a whole class of errors on real statements.

C. Ensembles (concept 5)

Q. Bagging vs Random Forest vs boosting?

Bagging trains trees on bootstrap samples and votes (variance reduction): 96.0% → 98.6%. Random Forest also randomises features per split, de-correlating the trees: 99.7%. Boosting trains sequentially on the residuals/weights of the previous model (bias reduction): HistGradientBoosting 99.9%, AdaBoost 80.1%.

Q. Why is AdaBoost weak here?

Depth-2 stumps on a ten-class problem need many rounds; SAMME re-weighting is sensitive to the noisy long-tail narrations, and it is the slowest model per unit of accuracy.

Q. What are the forest's top features?

is_debit, log_amount, amount, day, has_paise, month_end — the tokens a human would pick, which makes the forest easy to explain.

D. Regression (concept 1)

Q. What is the target and the baseline?

Next month's total spend of a customer from this month's panel row. Baseline: 'same as this month' — R^2 0.832, RMSE ₹13,590, because spend is sticky.

Q. Which models and which won?

Linear, Ridge, Lasso, degree-2 polynomial + Ridge, Decision Tree, Random Forest, Gradient Boosting. Gradient Boosting (400): CV R^2 0.875, test R^2 0.871, RMSE ₹11,881 — 13% below naive.

Q. What is GroupKFold and why?

k-fold cross-validation where all rows of one group (customer) go to the same fold. Without it a customer's January would train the model that predicts their February, inflating R^2 through identity leakage.

Q. Why are Ridge and Lasso equal to OLS?

4,400 rows comfortably support 34 coefficients, so the penalties barely bind. Ridge (L2) shrinks all coefficients; Lasso (L1) can zero some; at these alphas neither changes the fit.

Q. What drives the forecast?

Previous and current spend carry about 80% of the importance, then the month's income credits and the category mix. The calendar-month features add little with one year of data — each customer sees each season once — so multi-year histories are the first item in the future scope.

Q. RMSE vs MAE vs MAPE?

RMSE squares errors (large months dominate); MAE averages absolute errors; MAPE divides by the actual (we floor it at ₹500 to avoid division by near-zero months).

E. Clustering (concept 3)

Q. What features and how many segments?

13 continuous behavioural features per customer (median monthly spend, income, transactions, median debit, UPI/card/ATM/transfer shares, charges and refunds per month, average and minimum balance, spend–income ratio), standardised. Silhouette peaks at k = 5 for K-Means (0.329) and Ward (0.319).

Q. How do you know the clusters are real?

Two ways: K-Means and Ward agree (ARI 0.93), and both recover the generator's hidden archetypes with ARI 0.93 without ever seeing them.

Q. How exactly did you choose k?

K-Means silhouette for k = 2…8; the values within 0.005 of the maximum form a plateau (k = 5, 6) and I keep the smallest k on it — parsimony. The hidden archetype is never used for the choice; its ARI is reported afterwards as external validation.

Q. Why did you drop has_emi from the clustering features?

A standardised binary feature takes only two values about ±1 apart, so it dominates Euclidean distance and K-Means split every segment into EMI / no-EMI halves. With the 13 continuous features the segments match the behavioural types; has_emi is still reported per segment as a descriptor.

Q. Why medians in the customer profile?

One big-ticket month (fees, hospital) would otherwise define a customer's 'monthly spend' and 'average transaction'; the median month is robust to it.

Q. Name the segments.

High-value / business (53, mostly business); Low-activity (51, mostly low_activity); Cash-reliant (129, mostly pensioner); Digital-first (UPI heavy) (74, mostly student); Regular salaried (193, mostly salaried).

Q. K-Means vs Ward?

K-Means: iterative, assigns to nearest centroid, needs k, O(nk) fast. Ward: agglomerative, merges the pair that least increases within-cluster variance, gives a dendrogram, $O(n^2)$ memory. Silhouette, inertia (elbow) and ARI are the selection and validation tools.

Q. What is the silhouette?

(b − a)/max(a, b) per point: a = mean distance to own cluster, b = to the nearest other cluster; averaged, in [−1, 1]; higher is better separated.

F. Dimensionality reduction and LWR (concepts 4, 6)

Q. What did PCA show?

9 of 13 components for 95% variance — the profile features are only mildly correlated — yet a forest on the first 3 components recovers the archetype with F1 0.955 vs 0.981 on all features. Variance retained is not information retained.

Q. PCA vs Truncated SVD?

PCA centres the data then takes the top singular vectors — not possible on a sparse TF-IDF matrix without densifying it. Truncated SVD skips centring, so it works on sparse text: 100 components keep 87.6% of the variance and a logistic model on them reaches macro-F1 0.989 vs 0.999 on all features.

Q. Explain locally weighted regression.

For a query income x_0, weight every training point by $\exp(-(x-x_0)^2/2\tau^2)$, solve weighted least squares $\theta = (X^TWX)^{-1}X^TWy$ and predict $[1, x_0]\cdot\theta$. Non-parametric: no global parameters, every prediction re-fits (O(n)). Best τ here LWR τ=10.0: RMSE ₹12,476 vs ₹12,586 for the global line, because spend vs income bends.

Q. What does τ control?

Bias–variance: small τ follows local noise (wiggly), large τ converges to ordinary linear regression.

G. Application, deployment, business

Q. Describe the service pipeline.

rows → classify (deployed logistic model + direction mask; rule category alongside) → category mix → customer-month panel → customer profile → K-Means segment → next-month forecast (Gradient Boosting) with the naive guess → anomaly z-scores → insights → JSON.

Q. How are anomalies flagged?

Within each ML category, the robust z-score of log(amount): (x − median)/(1.4826·MAD); flagged above 3. Category-aware and explainable: '4× the typical ATM withdrawal'. Hints, never claims.

Q. Which endpoints?

GET /api/ml/status, /results, /figures/{png}; POST /api/ml/analyze (file → parsers → analysis), /analyze-json; GET /api/accounts/{id}/intelligence (session-protected, owner only).

Q. How does the UI fit the existing app?

Same hash router and screen/bind pattern, same i18n dictionary (Tamil/English keys intel.*), same card styles; an inline-SVG chart; the model card renders results/*.json and serves the figures from the API.

Q. What are the business insights?

Month-on-month trend; biggest category mover; largest outflow; charges found with a pointer to Findings; savings rate; segment; forecast beside the naive guess; unusual debits; count of model–rule disagreements.

Q. What would you not trust?

99% on synthetic data; forecasts from fewer than three months; segment of a very short statement; any output as a claim.

Q. Testing?

114 pytest tests: 101 from Vasool Raja plus 13 for the ML layer (generator determinism, masking, featurizer, panel/profile invariants, rule map, LWR exactness, service accuracy > 90%, endpoints, 415 on bad upload, 401/404 without session).

Q. What next?

Real consented statements via Account Aggregator with active learning on disagreements; multi-year data; confirmed findings as training labels; Tamil/Hindi tokens; Isolation Forest when labelled anomalies exist.

H. One-line definitions

Q. Stratified CV

Folds that preserve the class ratio.

Q. GroupKFold

Folds that keep all rows of a group together.

Q. Over-fitting

Train metric improves while validation metric falls.

Q. TF-IDF

Term frequency × inverse document frequency — rare informative tokens weigh more.

Q. Macro-F1

Mean of per-class F1, every class equal.

Q. Confusion matrix

Counts of true vs predicted classes.

Q. Adjusted Rand Index

Agreement between two partitions, chance-corrected.

Q. Explained variance ratio

Share of total variance a component carries.

Q. Silhouette

(b − a)/max(a, b) — cluster cohesion vs separation.

Q. MAD

Median absolute deviation — a robust spread.

Q. Bootstrap

Resampling rows with replacement.

Q. Complement NB

Naive Bayes estimated from each class's complement — robust to imbalance.

3-Minute Demo Script

1. Run `uvicorn api.main:app` (VASOOL_AUTH=off); open http://localhost:8000 → Intelligence.
2. Sample: hdfc priya salary 2026 → KPIs: debits, income, charges, segment, forecast vs naive.
3. Read the insights aloud: trend, biggest mover, charges → Findings, savings rate, segment, forecast, disagreements.
4. Sample: axis kavya longtail → 'show disagreements only': 30 of 41 lines — PAYROLL, BAJAJ FIN, SWIGGY CHENNAI, CSH WDL, QTRLY INT are generic transfers to the rules, the right class to the model.
5. Sample: canara amma pension → five charges; ₹50,000 college fee flagged as unusual.
6. Model card → tables live from results JSON → Fig 5, Fig 11, Fig 15.
7. Findings tab → rule engine's claims on the same statement: rules decide.

Numbers to Remember

Item	Value
Dataset	1,27,604 lines · 500 customers × 12 months · 10 categories · 5 archetypes · 35% long-tail narrations
Protocol	40k stratified sample → 80/20 · stratified 5-fold CV on train · GroupKFold(5) by customer for regression · test scored once
Classification	rules 71.5% · NB 93.4% · kNN 97.4% · DT 96.7% · LogReg 99.9% (macro-F1 0.997)
Ensembles	tree 96.0% · bagging 98.6% · RF 99.7% · GB 99.9% · AdaBoost 80.1%
Regression	naive R^2 0.832 / RMSE ₹13,590 → Gradient Boosting (400) R^2 0.871 / RMSE ₹11,881 (−13%)
Clustering	k = 5 · silhouette 0.329 · K-Means vs Ward ARI 0.93 · ARI vs hidden type 0.93
PCA / SVD	9 of 13 PCs for 95% · 3 PCs → archetype F1 0.955 · SVD-100 text F1 0.989
LWR	best LWR τ=10.0 · RMSE ₹12,476 vs linear ₹12,586
Engineering	114 tests · 6 endpoints · 7 notebooks · 15 figures · Docker · make train ≈ 25 min